

NanoGT Match Report: Nephropathic cystinosis

Disease: Nephropathic cystinosis (ORPHA:213)

Primary gene: CTNS

Gene CDS: 1104 bp

Inheritance: Autosomal recessive

Target tissues scored: kidney, multisystem

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: Vector does not naturally cover all annotated disease tissues: kidney, multisystem; No direct tissue overlap; treat this as weak precedent unless route or modality is changed; Cell-autonomous protein across multiple tissues; high transduction coverage may be required.

LYSOSOMAL MEMBRANE TRANSPORTER — TOP PRECEDENT

CAVEAT: The top-ranked precedent (Strimvelis, LV, ex vivo HSC) reflects lentiviral/HSC delivery class similarity, not mechanistic transferability. CTNS encodes cystinosin, a lysosomal **membrane transporter** — the same protein class as MCOLN1 (mucopolidosis type IV) and SLC17A5 (Salla disease). It cannot be secreted or taken up by neighbouring cells via the mannose-6-phosphate receptor. The Strimvelis match does not imply that ex vivo HSC gene addition is the appropriate strategy for cystinosis; the actual therapeutic requirement is per-cell delivery to renal tubular epithelium and other affected tissues. This score (6.4/10) should be read as “closest lentiviral/HSC precedent class” rather than “validated renal delivery precedent.” Same limitation as ML-IV — see LIMITATIONS_AND_FIXES.md Bug 1.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: Cystinosin lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs

Gene-addition compatibility: conditional

Preferred modality class: kidney or systemic gene addition

Evidence level/status: direct / needs_user_fact_check

Evidence summary: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues

Evidence source: OMIM CTNS 219800

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- No catalog program directly targets the annotated disease tissue(s); ranking is extrapolating from indirect precedents.
- No catalog vector naturally covers the annotated disease tissue(s); consider non-catalog vectors or alternative modalities.

- Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Strimvelis	LV	6.4/10	Medium	approved
2	BMN 307	AAV5	6.2/10	Medium	phase2
3	Libmeldy	LV	6.0/10	Medium	approved
4	OAV101-IT	AAV9	6.0/10	Medium	approved
5	Zolgensma	AAV9	6.0/10	Medium	approved

Match #1: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 6.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.50	1.0	Immunological protection of target tissue
Promoter availability	0.50	1.0	Validated tissue-specific promoters exist

Dimension	Score	Max	What it measures
Route of administration	0.50	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.43	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- No tissue overlap (disease: ['kidney', 'multisystem'], vector: ['hematopoietic'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Cystinosis lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues
- Mechanism source: OMIM CTNS 219800 (<https://omim.org/entry/219800>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-low privilege; renal immune surveillance is significant
- Promoter availability: Limited clinical-grade promoters validated for multisystem
- Route of administration: Delivery route to multisystem not well established in clinical programs
- ORGANELLE TARGETING: COMPATIBLE — CTNS standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: kidney, multisystem
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation

Match #2: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 6.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.50	1.0	Immunological protection of target tissue
Promoter availability	0.50	1.0	Validated tissue-specific promoters exist
Route of administration	0.50	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.24	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1353bp / cargo 4700bp (29% utilized)
- No tissue overlap (disease: ['kidney', 'multisystem'], vector: ['liver', 'lung', 'CNS'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Cystinosis lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints

- Mechanism evidence: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues
- Mechanism source: OMIM CTNS 219800 (<https://omim.org/entry/219800>)
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-low privilege; renal immune surveillance is significant
- Promoter availability: Limited clinical-grade promoters validated for multisystem
- Route of administration: Delivery route to multisystem not well established in clinical programs
- ORGANELLE TARGETING: COMPATIBLE — CTNS standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: kidney, multisystem
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent

Dimension	Score	Max	What it measures
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.50	1.0	Immunological protection of target tissue
Promoter availability	0.50	1.0	Validated tissue-specific promoters exist
Route of administration	0.50	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.95	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1521bp / cargo 8000bp (19% utilized)
- No tissue overlap (disease: ['kidney', 'multisystem'], vector: ['hematopoietic'])
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Cystinosis lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues
- Mechanism source: OMIM CTNS 219800 (<https://omim.org/entry/219800>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-low privilege; renal immune surveillance is significant
- Promoter availability: Limited clinical-grade promoters validated for multisystem
- Route of administration: Delivery route to multisystem not well established in clinical programs
- ORGANELLE TARGETING: COMPATIBLE — CTNS standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: kidney, multisystem
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation

Match #4: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 6.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.50	1.0	Immunological protection of target tissue
Promoter availability	0.50	1.0	Validated tissue-specific promoters exist
Route of administration	0.50	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.95	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)
- No tissue overlap (disease: ['kidney', 'multisystem'], vector: ['CNS', 'muscle', 'liver', 'heart'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Cystinosis lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues
- Mechanism source: OMIM CTNS 219800 (<https://omim.org/entry/219800>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-low privilege; renal immune surveillance is significant
- Promoter availability: Limited clinical-grade promoters validated for multisystem
- Route of administration: Delivery route to multisystem not well established in clinical programs
- ORGANELLE TARGETING: COMPATIBLE — CTNS standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

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- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #5: Zolgensma

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/motor neuron

Composite score: 6.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.50	1.0	Immunological protection of target tissue
Promoter availability	0.50	1.0	Validated tissue-specific promoters exist
Route of administration	0.50	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.95	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)
- No tissue overlap (disease: ['kidney', 'multisystem'], vector: ['CNS', 'muscle', 'liver', 'heart'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Cystinosin lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues
- Mechanism source: OMIM CTNS 219800 (<https://omim.org/entry/219800>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must in-

dividually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose

- Immune privilege: moderate-low privilege; renal immune surveillance is significant
- Promoter availability: Limited clinical-grade promoters validated for multisystem
- Route of administration: Delivery route to multisystem not well established in clinical programs
- ORGANELLE TARGETING: COMPATIBLE — CTNS standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

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- Vector does not naturally cover all annotated disease tissues: kidney, multisystem
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- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
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